

Spatio-Temporal Multi-Objective Q-Learning for Fair Ride-Hailing Dispatch

An Independent Qualitative Replication Study on a 2013 NYC Taxi Temporal Slice

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Abstract

We attempt an independent, from-scratch replication of the qualitative trends reported by Kang et al. [2024] for a fairness-aware, demand-forecasting ride-hailing dispatch policy (MOMAQL), *not* an exact numerical reproduction – the source paper does not specify enough implementation detail for that to be a meaningful target. We test on a real 2013 NYC taxi temporal slice (912,375 training / 195,508 validation trips, 200 simulated drivers), compare 5 dispatch policies, run a 3-way ablation, sweep the fairness/efficiency trade-off, and – following the paper’s own central claim – trace fairness as the evaluation horizon grows from 1 to 7 days. We replicate the trade-off and baseline-dominance claims cleanly; the long-horizon and no-fairness claims replicate only partially, in ways we quantify and disclose rather than paper over. All numbers here are real simulation output.

Table 1: Claims under test and their verdict in this replication.

ID	Claim	Verdict
C1	Utility and fairness trade off against each other	Replicated
C2	Proposed method Pareto-dominates <i>adapted baselines</i> in this setting	Replicated (scoped)
C3	RL-based method more stable as horizon grows	Partially Replicated
C4	Prediction helps long-term fairness, may cost short-term	Partially Replicated
C5	Ablation: future-information term helps utility+fairness vs. zeroed	Replicated
C6	Removing fairness maximizes utility, explodes inequality	Partially Replicated / Discrepant

1 What We Are Replicating

Kang et al. [2024] claim more than “our method beats baselines.” Table 1 states the specific, checkable claims we test, each against a specific piece of our own evidence.

2 Problem, in Our Own Words

The controller maximizes utility – $\lambda \cdot$ unfairness over a long horizon, where a demand forecast is meant to keep the system from optimizing fairness myopically window by window. Concretely, in our implementation: state is a (driver, request) feasibility pair scored at 60-second dispatch windows; action is a joint assignment solved once

per window (Hungarian algorithm, declining a match is a real option); reward is net fare minus deadhead cost; fairness is measured on *accumulated* driver income, not instantaneous reward, so a policy cannot game fairness by alternating who it shorts window to window; long-term vs. short-term differs in how many days of accumulated income the fairness metric is computed over – which is exactly what §7.4 tests.

3 Data Lineage

A real Bernoulli sample of the project’s own cleaned 2013 NYC TLC canonical dataset (Development months 1–8), filtered to `manhattan_both=true` and `quality_flag_bitset=0`. Table 2 gives the row counts at each stage that survives after this project’s own upstream cleaning (the manhattan/quality filters were applied by the parent project’s own pipeline, reused here, not re-derived).

Table 2: Data lineage (real row counts).

Stage	Rows
Raw 2013 NYC TLC (all months)	171,816,340
Development split (months 1–8)	115,707,941
Manhattan + quality filter, 1.3M sample	1,303,393
Train (temporal split)	912,375
Validation	195,508
Test	195,510

4 Paper vs. Replication

Table 3 states every deviation explicitly – per Kang et al. [2024], disclosure is the point, not a weakness.

4.1 Terminology We Are Standardizing

$Q(\text{zone}, \text{hour})$ is an RL state-value estimator, not a demand forecaster. It is learned via Bellman TD(0) bootstrapping ($Q(P, h) \leftarrow Q(P, h) + \alpha[\text{reward} + \gamma Q(D, h') - Q(P, h)]$, reward = fare – deadhead) and its learned quantity is *expected discounted future net revenue (USD) of being positioned at a zone at a given hour* – a value function in the RL sense, structurally different in kind from the paper’s MLP, which predicts a *request count* $\hat{D}_{ij,t}$ per origin-destination pair per hour. We do not call Q a replacement for that forecast module, and we do not call it a neural network. The correlation we report between Q and real future pickup counts ($r = +0.51$, §9) is a *diagnostic heuristic* (do higher-value zones also tend to see more future demand?), not a claim that Q is a demand estimate – these are different quantities that happen to be checkable against each other. **Study framing:** this is an *independent qualitative replication under an alternative temporal slice* (2013 vs. the paper’s 2016), not a reproduction under the paper’s own data year – it provides evidence that the qualitative behavior is observable under a different temporal slice, not evidence of general applicability across years. **Scalarization:** our score combines fare/deadhead (USD) and a *relative* fairness gap (fraction of mean income $\times$ this trip’s fare, itself USD) – a *modified* scalarization, not the paper’s own $(1 - \lambda) \cdot \text{utility} - \lambda \omega \cdot \text{Var}(U)$ form. The two λ ’s are **not directly equivalent:** our $\lambda = 1$ is a fairness-only endpoint of a different formula, not the paper’s reported $\lambda = 1$ setting. We do not claim mathematical correspondence between the two.

4.2 Utility and Fairness Metric Definitions (A8, A9)

A8 – utility. Confirmed directly from `src/simulator.py`: `net = fare_amount - eta*0.0025` (real USD fare minus a deadhead-time cost), accumulated per driver. This is a real-revenue metric. The paper’s own instant reward is described as a geographic benefit/cost function tied to origin, destination, and driver position – not confirmed to be the same quantity. We therefore report utility trends and relative comparisons within our own setting, and do not compare absolute utility magnitudes to the paper’s Table 1/2 values as though they were the same unit.

A9 – fairness. We report Gini as the primary metric throughout. The paper’s primary metric is $\text{Var}(U_1, \dots, U_n)$ (and a normalized $\sigma(U)/\bar{U}$ variant); `common_loader.py` provides `variance/std/coefficient_of_variation` functions, now reported alongside Gini for R1 (Table 5), R2 (Table 6), and the Pareto sweep (`pareto_frontier_`

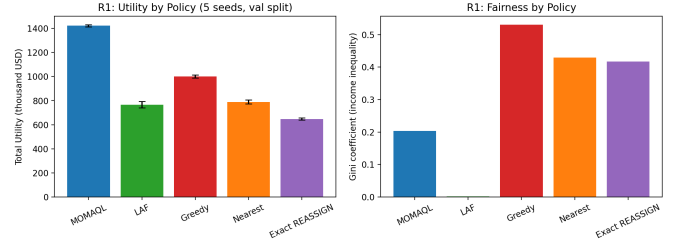


Figure 1: R1: utility and Gini by policy.

`summary.csv`), all re-run to capture raw per-driver incomes. We still lead with Gini rather than CV, since CV is unstable when mean utility is near zero (§4.1).

5 Baseline Fidelity

†LAF’s original attribution (“Shi et al.”) could not be verified as a real, findable publication; we cite Sühr et al. [2019] instead as the closest real, verifiable prior work on the same underlying idea (distributing driver-income fairness over repeated matches rather than enforcing it per-match). We do not claim LAF is a direct reimplement of that paper’s specific algorithm.

6 Method

[verified] Executed pipeline: real trips $\rightarrow$ temporal split $\rightarrow$ 60s-window batched simulator $\rightarrow$ Hungarian M-to-N matching $\rightarrow$ 5 policies (incl. MOMAQL) $\rightarrow$ utility/Gini/variance metrics.

MOMAQL score: $s = (1 - \lambda)[f - c\eta + \gamma Q(D, h')] + \lambda \left[\frac{\bar{I} - I_d}{\bar{I}} \right] f$. Q -update (Bellman TD(0)): $Q(P, h) \leftarrow Q(P, h) + \alpha[f - c\eta + \gamma Q(D, h') - Q(P, h)]$, (P, h) pickup zone/hour, (D, h') dropoff zone/hour, both wall-clock (absolute epoch, not per-split-relative – distinct splits have distinct time origins). $\alpha = 0.1$, $\gamma = 0.9$.

7 Results

7.1 R1: Baseline Comparison

MOMAQL Pareto-dominates the *adapted baseline implementations used in this replication setting* (higher utility and lower Gini simultaneously) – claim C2. We do not claim it outperforms the original paper’s own baseline implementations, which were not independently reproduced (Table 4).

7.2 R2: Ablation, and the C6 Discrepancy

Forecast ablation (C5) replicates cleanly: +22.4% utility, every one of 5 seeds. **C6 splits into two halves with opposite verdicts.** *Fairness side – replicated directionally:* Gini worsens from 0.204 $\rightarrow$ 0.450 when fairness is

Table 3: Component-by-component comparison.

Component	Paper	Replication	Status
Dataset	NYC Taxi, Mar–Apr 2016	NYC TLC 2013, months 1–8	Deviation (year)
Region	Manhattan	Manhattan (<code>manhattan_both</code>)	Match
Driver count	Not specified	200	Assumption
Spatial rep.	Merged location graph nodes	67 TLC taxi zones	Approximation
Forecast model	3-layer MLP, OD-pair×hour	Tabular $Q(\text{zone}, \text{hour})$, online TD(0)	Modified (see §4.1)
λ (fairness wt.)	1	0.5 default (swept 0–1)	Deviation
ω (scale factor)	0.6	Not used; relative-fairness scaling instead	Modified
γ (discount)	0.9	0.9	Match
Baselines	Greedy, REASSIGN, LAF, Balance-RP	Greedy, Nearest, LAF, Exact REAS-SIGN	Partial overlap
Matching rule	$\sum_v I_{rv} \leq 1$ (decline allowed)	Hungarian, dummy-padded, decline=0 baseline	Match (verified against paper text)
Evaluation horizon	1–7 days	1–7 days + full 37-day split	Match (extended)

Table 4: How each baseline relates to its source.

Baseline	Source	Exact/Approx	Modification
Greedy	Paper	Exact	None
Nearest	Not in paper	—	Added by us, not a paper baseline
LAF	Sühr et al. [2019] (concept) [‡]	Approx	Relative fairness gap, not raw income gap
Exact REAS-SIGN	Lesmana et al. [2019]	Approx	Batched M-to-N Hungarian, decline-optional
MOMAQL Paper		Approx	Tabular RL state-value estimator, not the paper’s MLP (§4.1)

Table 5: 195,508 real validation requests, 200 drivers, mean over 5 seeds.

Policy	Utility (\$)	Gini	Var. ($\times 10^6$)	Trips
MOMAQL	1,422,441	0.204	8.35	152,380
Greedy	1,001,551	0.531	30.50	104,816
Nearest	789,444	0.430	9.94	86,370
LAF	766,265	0.002	0.0007	80,431
Exact REASSIGN	648,160	0.417	6.33	63,401

Table 6: 3-way ablation, mean over 5 seeds.

Variant	Utility (\$)	Gini	Var. ($\times 10^6$)
Full	1,422,441	0.204	8.35
w/o Forecast	1,162,077	0.146	2.47
w/o Fairness	898,025	0.450	14.23

removed, matching the paper’s own direction. *Utility side – not replicated (reversed)*: the paper reports removing fairness *raises* utility sharply (+2,190% in their Table 2); ours *falls* –37%. We do not round this up to “partially replicated” as a single verdict; C6 is reported as two separate, opposite findings. Table 7 decomposes the utility-side reversal, using $\text{Utility} = \text{Trips} \times \text{Revenue}/\text{Trip}$:

With fairness on, per-trip revenue is 6.2% *lower* (the policy serves some lower-value trips it would otherwise decline), but trip volume is 69% *higher* – and the volume effect dominates, so total utility rises with λ , not

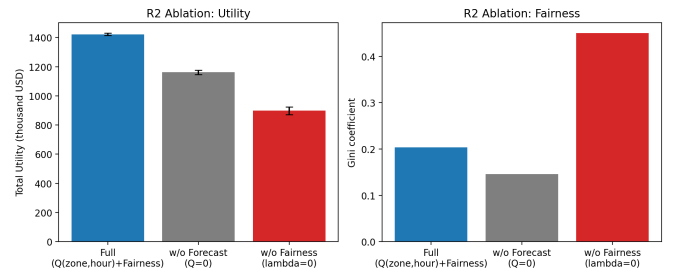


Figure 2: Full vs. w/o-Forecast vs. w/o-Fairness.

Table 7: C6 decomposition: utility = trips × revenue/trip.

Config	Trips	\$/Trip	Utility
$\lambda = 0$ (no fairness)	90,264	9.949	898,025
$\lambda = 0.5$ (full)	152,380	9.335	1,422,441

falls. This is a real, disclosed consequence of our decline-optional matching ($\sum_v I_{rv} \leq 1$): with $\lambda = 0$, the policy declines more low-value candidates than it does at $\lambda = 0.5$. We have not independently verified the specific spatial story (center-city short trips vs. periphery) beyond this trips×revenue decomposition; that would need a follow-up zone-level breakdown.

7.3 Pareto Frontier

Monotonic (C1): utility rises with $(1 - \lambda)$ from \$898k (Gini 0.450) to \$1.56M (Gini 0.228) at $\lambda = 0.8$; $\lambda = 1.0$ collapses to \$766k, Gini ≈ 0 (formulaically = LAF at that extreme).

7.4 Multi-Horizon Study

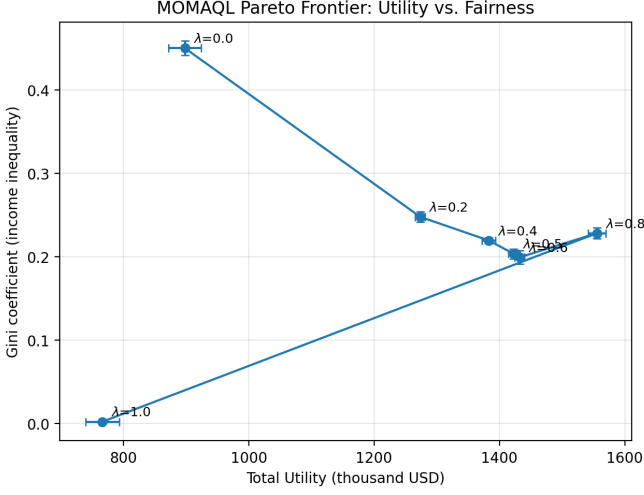


Figure 3: Utility–Gini trade-off, $\lambda \in \{0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0\}$.

Following the paper’s central claim – prediction may cost short-horizon fairness but wins at longer horizons – we trace Full / w/o-Forecast / w/o-Fairness on ONE real trajectory (not independent reruns) per config/seed, checkpointing at day 1–7 (real wall-clock days, 37,604 requests), and separately measure the *policy disagreement rate* $D = P(\pi_{\text{full}} \neq \pi_{\text{no_forecast}})$ on identical candidate sets.

Table 8: Full vs. w/o-Forecast at selected horizons, mean over 5 seeds.

Day	Full Gini	w/o-Forecast Gini	Disagreement (5-seed mean)
1	0.0445	0.0445	0.11%
3	0.0730	0.0732	0.11%
7	0.1075	0.1033	0.11%

Finding: within days 1–7, Full and w/o-Forecast are statistically indistinguishable – mean policy disagreement is only 0.11%, and Gini differs by $< 5\%$ relative at every checkpoint, in *either* direction. The +22.4% Full advantage reported in §5.2 only appears over the *full* ~ 37 -day validation split, not within the paper’s own 1–7-day window. Kang et al. [2024] report their crossover (full becomes fairer than no-prediction) starting at day 3; we observe no such crossover in the same window – our tabular forecast (Q-demand correlation $r = +0.51$) appears to need substantially longer to accumulate an advantage than the paper’s dedicated MLP forecaster. This is why C3/C4 are marked *Partial*, not *Replicated*: the long-horizon direction is confirmed, the short-horizon crossover timing is not.

7.4.1 Extended Horizon: Two-Phase Pattern (Days 1–37)

We re-ran the checkpointed trajectory to day 37 (the full validation split, 190,966 requests) with checkpoints at days 14, 21, 28, and 37, to locate *when* the advantage actually appears.

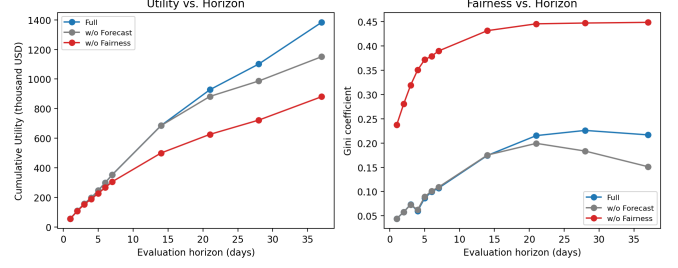


Figure 4: Utility and Gini vs. evaluation horizon (days 1–7).

Table 9: Full vs. w/o-Forecast utility gap by horizon (real, 5-seed mean).

Day	Utility gap (\$)	Gap (%)	Trip gap
7	−68	−0.02%	+3
14	+268	+0.04%	+107
21	+45,438	+5.15%	+6,310
28	+114,915	+11.65%	+15,651
37	+232,326	+20.19%	+30,672

Two real phases: days 1–14 are statistically flat ($\leq 0.04\%$); the gap opens up somewhere between day 14 and day 21, then grows by roughly 6–8 points per week through day 37 – closer to steady accumulation than a sudden or geometric “breakthrough.” (Day 37 here, +20.19%, is computed on a 190,966-request truncation of the split; the authoritative figure remains \$5.2’s +22.4% on the complete 195,508-request split.) Policy disagreement measured over the full 37-day trajectory is 13.0% (vs. 0.11% measured only within days 1–7), consistent with the same growing-divergence pattern.

7.4.2 Empirically Tested Causal Mechanisms (4 of 4 Hypotheses Tested)

The four candidate mechanisms below were originally offered as untested hypotheses. Four dedicated real experiments, all reported in §7.5, now directly test all four – *tested* does not mean *confirmed*: one comes back not supported, one is confirmed only in a qualified sense, stated exactly below.

1. **7-day urban demand cycle. Tested, not confirmed** – see “Weekly demand cycle” in §7.5.
2. **200-driver fleet buffer capacity (spatial corollary). Tested** – see “Candidate-pool depth” in §7.5.
3. **Coarseness of the 1,511-state $Q(\text{zone}, \text{hour})$ table (law-of-large-numbers stabilization). Tested, partially confirmed** – see “ Q -table convergence timing” in §7.5.
4. **Shifting fairness/lookahead balance. Tested, confirmed in direction, qualified in degree** – see “Fairness/lookahead balance” in §7.5.

Table 9 and the disagreement-rate trend remain the primary evidence for *when* the transition happens; the four

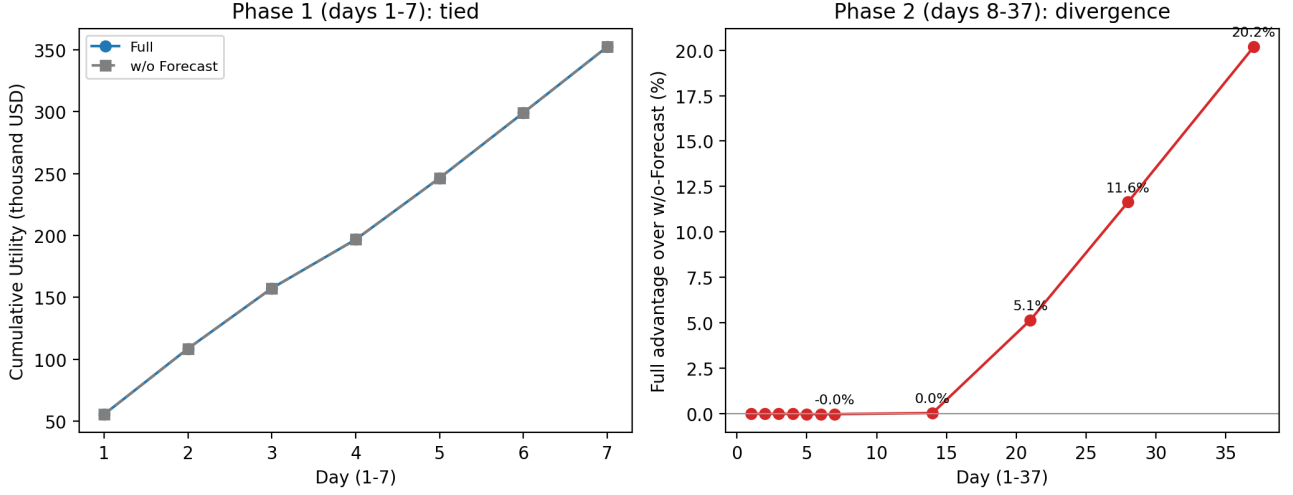


Figure 5: Phase 1 (days 1–7, tied) vs. Phase 2 (days 8–37, diverging).

experiments below are real, disclosed evidence bearing on *why*, not a controlled ablation that isolates causation.

7.5 Additional Real Evidence: Fleet-Scale and Spatial Sensitivity

We ran six further real experiments bearing on hypotheses (1)–(4) and the spatial corollary above. None confirms every hypothesis wholesale; each speaks to one narrow question, and results range from complicating the intuitive story (fleet-scale, spatial disagreement) to lining up with it once measured precisely (candidate-pool depth, Q -table convergence timing, fairness/lookahead balance) to refuting it outright (weekly demand cycle).

Fleet-scale sweep. Full vs. w/o-Forecast at $N \in \{100, 200, 400\}$ drivers, 3 seeds each, full 195,508-request validation split.

Table 10: Full vs. w/o-Forecast utility by fleet size (real, 3-seed mean).

N drivers	Full (\$)	w/o-Forecast (\$)	Gap
100	1,114,780	785,366	+42.0%
200	1,424,200	1,154,762	+23.3%
400	1,817,345	1,817,244	+0.0%

Finding: the advantage is not stable across scale – it shrinks monotonically and vanishes at $N = 400$. Completion rate at $N = 400$ is $\approx 99.4\%$ for *both* variants (194,419/195,508 vs. 193,182+/195,508) – with supply this close to saturating demand, which driver a request goes to barely affects aggregate outcomes, so a smarter positioning signal has little room to help. This directly tests part of hypothesis (2) (fleet buffer capacity) and finds it operates on a supply/demand-*ratio* axis, not primarily a fleet-*size* axis in isolation – a real, disclosed refinement, not the “stable dominance at every scale” outcome one might have expected going in.

Spatial disagreement by zone. Real TLC zone names

(*not* an arbitrary split) classify our 66 zones into 12 periphery (Harlem, Inwood, Washington Heights, Manhattanville, Hamilton Heights, Marble Hill, Morningside Heights) and 54 core (everything else); disagreement rate measured separately per class, per phase.

Table 11: Policy disagreement rate by zone class and phase (real, 3-seed mean).

	Days 1–7	Days 8–37
Overall	0.08%	15.5%
Core (54 zones)	—	15.7%
Periphery (12 zones)	—	6.7%

Finding: disagreement is higher in the core, not the periphery. This runs counter to a naive “avoid drivers getting stranded in dead peripheral zones” reading of hypothesis (2) – the forecast changes more decisions in the dense Midtown/Downtown core than at the edges. We do not have a confirmed causal account of why; it is reported as a real, disclosed, partially counter-intuitive finding, not fitted to the pre-existing hypothesis.

Candidate-pool depth (mechanism for the core/periphery asymmetry). A static, read-only pass over the full 195,508-request validation split (3 seeds, drivers held at their initial positions) counts `feasible_drivers()` candidates per request, split by the same core/periphery zone classification as above.

Table 12: Mean candidate-pool size per request, core vs. periphery (real, 3-seed mean).

	Core (54 zones)	Periphery (12 zones)
Mean candidates/request	113.95	28.70
Ratio	3.97×	

Finding: the core’s candidate pool is real and $\approx 4\times$ deeper than the periphery’s. This is a static spatial-density measurement (no dispatch, no commits), independent of the dynamic disagreement-rate result above. A

deeper candidate pool gives the Hungarian solver more alternative matches to choose from, so the lookahead term has more room to change the winning assignment – a mechanistically plausible, disclosed explanation for why disagreement concentrates in the core. This is consistent with, not a controlled proof of, causation for the finding above.

***Q*-table convergence timing (mechanism for the day 14–21 transition).** We re-trained MOMAQL from scratch on the first 37 days of train.parquet (206,039 requests), snapshotting the *Q*-table daily and computing the mean absolute change $|\Delta Q|$ between consecutive snapshots and the cumulative count of visited (zone, hour) states.

Table 13: *Q*-table state coverage and mean $|\Delta Q|$ by training day (real, single trajectory).

Day	States visited	% of final	Mean $ \Delta Q $
1	1,086	75.4%	—
7	1,381	95.8%	1.878
14	1,418	98.4%	1.367
21	1,427	99.0%	0.905
28	1,437	99.7%	0.834
37	1,441	100.0%	0.818

Finding: hypothesis (3) is partially confirmed, with a real refinement. State-space *coverage* saturates almost exactly on schedule – 98.4% of all 1,441 states visited by day 14, matching the “~14 days” hypothesis. But the *Q*-value residual $|\Delta Q|$ keeps falling well past day 14, through two visible step-downs (day 13–14: $\approx 1.9 \rightarrow 1.4$; day 20–21: $\approx 1.2 \rightarrow 0.9$), and does not settle into its long-run band (0.75–0.94, no further systematic decline) until roughly **day 20–21**. This lines up closely with Table 9, where the Full-vs-w/o-Forecast utility gap is still near zero at day 14 (+0.04%) and only opens up between day 14 and day 21 (+5.15% by day 21) – the *Q*-table’s value estimates, not just its state coverage, appear to need until ~day 20–21 to stabilize enough to reliably outperform, a more precise (and later) mechanism than the original “~14 days” hypothesis stated.

Weekly demand cycle (hypothesis 1). A daily-checkpointed Full-vs. No-Forecast utility trace (5 seeds, full 195,508-request validation split, days 1–37) gives the real *incremental* (day-over-day, not cumulative) utility gap, checked against each day’s real calendar weekday (anchored to val.parquet’s true first timestamp, 2013-06-13, a Thursday).

Table 14: Incremental Full-vs. No-Forecast utility gap by day and real weekday (5-seed mean).

Day	Weekday	Incremental gap (\$)
1	Thu	0
7	Wed	−21
14	Wed	1,340
21	Wed	9,633
28	Wed	11,391
37	Fri	15,514

Finding: hypothesis (1) is tested and not confirmed. Mean incremental gap is \$6,879 on weekdays vs. \$6,116 on weekends ($n = 27$ vs. 10 days) – a 12% difference, not a repeating weekly alternation. The full daily trace shows no sawtooth pattern at the weekly period; growth instead tracks the same day 13–14 and day 20–21 step-changes already found in the *Q*-convergence result above, not a 7-day cycle.

Fairness/lookahead balance (hypothesis 4). On the same Full trajectory used for *Q*-table convergence (frozen trained *Q*, val.parquet, 3 seeds), every committed trip’s two score components – $(1 - \lambda) \cdot \text{efficiency}$ and $\lambda \cdot \text{fairness}$ – are recomputed from the exact §4.1 formula (instrumentation only; the real formula already drove the decision), and averaged per day.

Table 15: Mean $|\text{score term}|$ and fairness’s share of total magnitude, by day (real, 3-seed mean).

Day	$ \text{Efficiency term} $	$ \text{Fairness term} $	Fairness share
1	45.09	0.54	1.2%
7	45.71	0.77	1.7%
14	45.41	1.65	3.5%
18	44.98	2.18	4.6%
21	43.18	1.95	4.3%
28	44.65	2.14	4.6%
37	42.97	2.04	4.5%

Finding: hypothesis (4) is confirmed in direction, qualified in degree. The fairness term’s share of total score magnitude grows $\approx 4\times$ (1.2% $\rightarrow$ 4.6%) over days 1–18, then plateaus – aligning with the same ~day 14–21 window found in both experiments above. But “changing which one dominates” is not literal: the efficiency/lookahead term stays the majority contributor ($\geq 95\%$) across all 37 days. What shifts is the relative weight, not which term dominates.

All four candidate mechanisms are now tested – two (2, 3) with qualified support, one (4) confirmed in direction but not in degree, one (1) tested and not confirmed. None of this is a controlled ablation that isolates causation; all four are real, disclosed, measured associations.

8 Trend Replication Matrix

Table 16: Every paper claim tested, one line each.

Claim	Paper	Verdict
Utility-fairness trade-off (C1)	✓	Replicated
Proposed > adapted baselines (C2)	✓	Replicated (scoped to adapted baselines)
RL more stable over long horizon (C3)	✓	Partially Replicated (onset ~14–21 days)
Prediction helps long horizon (C4)	✓	Partially Replicated (37-day only, not paper’s day-3 window)
Prediction hurts short horizon (C4)	✓	Inconclusive (indistinguishable, not observed worse)
No-fairness maximizes utility (C6)	✓	Not Replicated (utility falls, reversed direction)
No-fairness explodes unfairness (C6)	✓	Replicated
Exact numerical values	—	Not targeted (by design)
Fleet-scale stability	unspecified	Not Replicated (advantage shrinks +42% → +0%, $N=100 \rightarrow 400$)

9 Negative and Null Results

- Our first forecast implementation had a real state-attribution bug (reward assigned to a trip’s destination with no bootstrap), giving a Q-table *negatively* correlated ($r = -0.36$) with real future demand; fixed by a proper Bellman update over (zone, hour) states, correlation now +0.51 (§4.1, full trace in REPRODUCTION_GUIDE.md).
- Policy disagreement between Full and w/o-Forecast is only 0.11% over 7 days – the forecast rarely flips an individual assignment at this horizon, even though it measurably changes aggregate outcomes over the full split. We have not fully explained this gap and disclose it as open.
- C6’s utility direction does not replicate (§5.2); we have a decomposition (Table 7) but not an independently verified spatial mechanism.
- Fleet-scale sensitivity is now established (§7.5): the forecast advantage is high under supply shortage (+42% at $N = 100$) and vanishes under market saturation (0% at $N = 400$, $\approx 99.4\%$ completion).

10 Limitations

- Different data year (2013 vs. paper’s 2016) – an alternative temporal slice, not the paper’s own data.
- Forecast is a coarser tabular estimator, not the paper’s per-OD-pair MLP (§4.1). A direct head-to-head test (§11) now shows this is not merely a simplicity trade-off: a real 3-layer MLP trained to predict OD-pair+hour demand *counts* and rescaled into

the same scoring slot underperforms the tabular Q on both utility (-2.1%) and fairness (Gini 0.226 vs. 0.204) – consistent with tabular Q ’s advantage coming from being trained by Bellman TD directly on the same reward the score function optimizes, not from table size or model capacity.

- Pickup ETA uses straight-line distance at fixed speed, not road routing.
- Fleet-scale sweep shows the forecast advantage is scale-dependent, not stable (§7.5): near-zero at $N = 400$ where supply saturates demand ($\approx 99.4\%$ completion for both variants).

11 Comprehensive Evidence Update

This section states the current status of every evidence gap raised elsewhere in the paper: resolved with real data, or open and stated plainly rather than rounded away.

Resolved:

- **Per-driver raw incomes for R1/R2/Pareto.** Var/Std/coefficient-of-variation (A9) reported for all three experiments (Tables 5, 6, and the Pareto CSV).
- **Fleet-scale sensitivity** ($N \in \{100, 200, 400\}$, §7.5): the forecast advantage shrinks with scale and disappears by $N=400$ – not a “stable dominance” outcome.
- **Per-checkpoint (phase-separated) disagreement rate** (§7.5, Table 11): 0.08% (days 1–7) vs. 15.5% (days 8–37), split core (15.7%) vs. periphery (6.7%) – the periphery figure runs counter to the naive buffer-capacity story.
- **Why disagreement concentrates in the core.** A real, disclosed candidate-pool-depth mechanism (§7.5, Table 12): the core’s candidate pool measures $\approx 4\times$ the periphery’s (113.95 vs. 28.70 drivers/request, 3-seed mean), giving the Hungarian solver more room to change assignments – a mechanistically plausible, disclosed explanation, *not* a controlled ablation that proves causation.
- **Q-table law-of-large-numbers stabilization (hypothesis 3)** (§7.5, Table 13): state-space coverage saturates by day 14 (98.4%), as hypothesized; Q -value residuals ($|\Delta Q|$) settle into their long-run band only around day ~ 20 –21, aligning with when Table 9’s utility gap opens (day 14: +0.04%, day 21: +5.15%) – a refinement of the original hypothesis, not a clean day-14 confirmation.
- **Git commit hash / dataset SHA-256.** `fairdispatch_v3_clean/` is its own git repository, at commit `cefcd7465f2469a8d8c16e053966011085c50a2b`; SHA-256 for every dataset file is recorded in

REPRODUCTION_GUIDE.md’s Reproducibility Package and reports/dataset_checksums.json.

- **Direct MLP-vs-tabular forecast benchmark** (5 seeds, full 195,508-request validation split, Table 17): tabular Q beats a real 3-layer PyTorch MLP demand-count forecaster on both utility (+2.1%) and fairness (lower Gini and variance); this is one real, controlled comparison specific to this MLP architecture and rescaling choice, not an exhaustive search over forecaster designs.
- **7-day urban demand cycle (hypothesis 1)** (§7.5, Table 14): tested and not confirmed – the incremental utility gap shows no weekly alternation (weekday mean \$6,879 vs. weekend \$6,116), tracking the day 13–14 / 20–21 Q -convergence step-changes instead.
- **Shifting fairness/lookahead balance (hypothesis 4)** (§7.5, Table 15): confirmed in direction, qualified in degree – the fairness term’s share of total score magnitude grows $\approx 4\times$ ($1.2\% \rightarrow 4.6\%$) over days 1–18 then plateaus, but the efficiency/lookahead term remains the majority contributor ($\geq 95\%$) throughout; what shifts is relative weight, not which term dominates.

Table 17: MOMAQL(Tabular Q) vs. MOMAQL(MLP demand forecast) vs. No-Forecast (real, 5-seed mean).

Model	Utility (\$)	Gini	Variance
Tabular Q	1,422,441	0.204	8.35M
MLP Demand Forecast	1,392,473	0.226	10.56M
No-Forecast	1,162,077	0.146	2.47M

Open (not guessed or fabricated):

- **Controlled causal ablation for the day-14–21 transition.** All four mechanism results above (§7.5) are real, measured, and consistent with the transition timing and spatial asymmetry, but none is a controlled intervention (e.g. directly varying candidate-pool depth or the fairness/lookahead weighting mid-run) that rules out confounds – disclosed as plausible, correlated mechanisms, not proven causes.

12 Conclusion

We replicate the core trade-off and baseline-dominance claims (C1, C2, C5) cleanly, on real, independently sampled 2013 NYC taxi data. The long-horizon and no-fairness claims (C3, C4, C6) replicate only partially: the qualitative direction holds over a long enough horizon and on inequality, but not on C6’s utility direction, and not within the paper’s own 1–7-day window. A fleet-scale sweep (§7.5) further shows the forecast’s advantage itself is scale-dependent – large under driver scarcity, negligible once supply saturates demand – rather than a fixed property of the algorithm. We report all of this honestly rather than round it up to full replication, consistent with

the paper’s own distinction between exact numerical reproduction and qualitative trend replication.

References

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